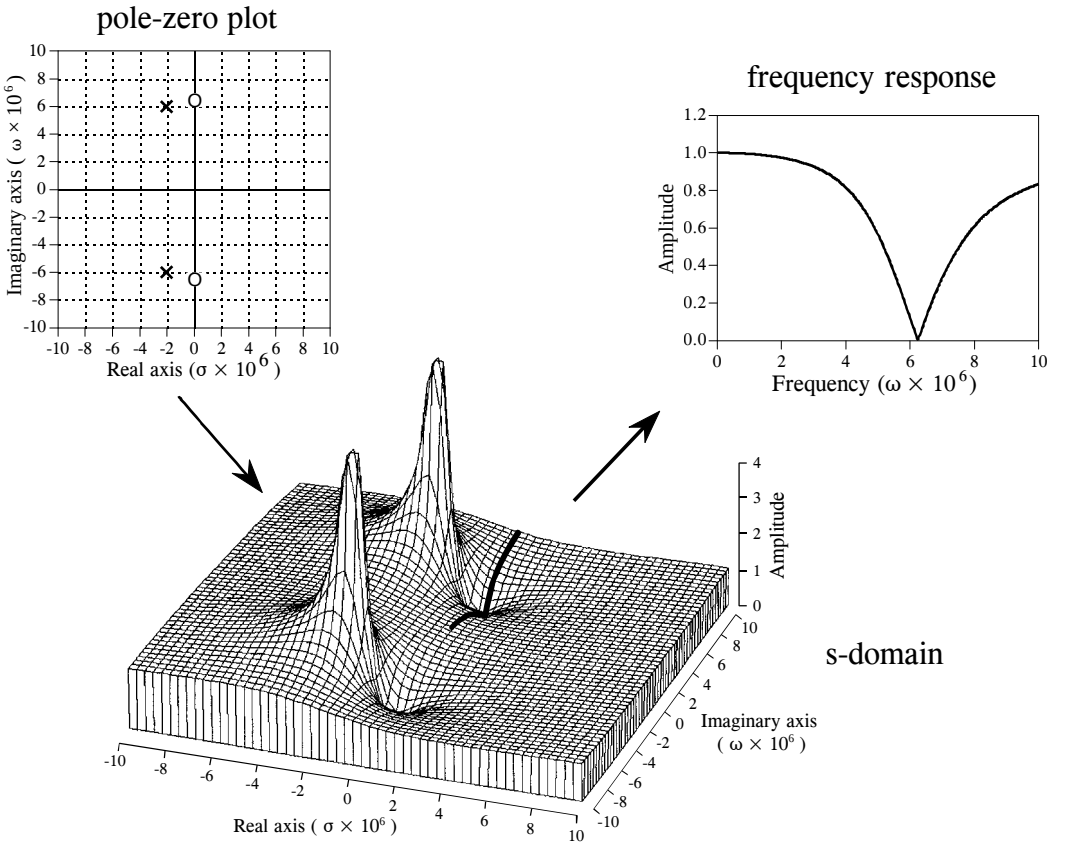


FIGURE 32-7
Poles and zeros in the s-domain. These illustrations show the relationship between the pole-zero plot, the s-domain, and the frequency response. The notch filter component values used in these graphs are: R=220 Ω, C=470 ρF, and L = 54 μH. These values place the center of the notch at ω = 6.277 million, i.e., a frequency of approximately 1 MHz.

The Importance of Poles and Zeros

To make this less abstract, we will use actual component values for the notch filter we just analyzed: $R = 220\Omega$, $L = 54\,\mu H$, $C = 470\,\rho F$. Plugging these values into the above equations, places the poles and zeros at:

$$z_1 = 0 + j\,6.277\times10^6 \qquad p_1 = -2.037\times10^6 + j\,5.937\times10^6$$

$$z_2 = 0 - j\,6.277\times10^6 \qquad p_2 = -2.037\times10^6 - j\,5.937\times10^6$$

These pole and zero locations are shown in Fig. 32-7. Each zero is represented by a circle, while each pole is represented by a cross. This is called a **pole-zero diagram**, and is the most common way that s-domain data are displayed. Figure 32-7 also shows a topographical display of the s-plane. For simplicity, only the magnitude is shown, but don't forget that